Reproducibility Summary

Replication package for "Dynamic Joint Calibration of Combined VaR and ES Forecasts: Evidence from US Industries and the Yen"

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This document summarizes the replication package of the paper: what it contains, what is needed to run it, how each table and figure is produced and what the author's reproduction check found. Its structure follows the template README of the Social Science Data Editors (Vilhuber et al., 2022). The data are described in the companion document *Data Sources*. The package is available at https://github.com/sheyung21-crypto/var-es-dynamic-joint-calibration.

1. Overview

The package contains the Python code, the settings fixed before the evaluation and the archived results of the paper. The code estimates ten base models for value-at-risk (VaR) and expected shortfall (ES), combines their forecasts, applies the calibration methods compared in the paper, and computes the scores, backtests and bootstrap tests. Separate programs then produce the 13 tables and 5 figures of the paper.

The paper reports two studies. Study 1 uses 30 US industry portfolios and six international equity indices. Study 2 uses the yen–US dollar exchange rate from 1985 to 2009 as an out-of-sample test. The settings of both studies were fixed before the evaluation data were analyzed (Study 1) or downloaded (Study 2); `provenance/Prespecified_Design.md` gives the version-control records.

On the computer described in Section 3.3, a full rerun of Study 1 takes about 26 minutes, a rerun of Study 2 about 2.4 minutes, and rebuilding all tables and figures from saved results about 21 seconds.

2. Data Availability

All data come from public providers. Only the yen series can be included in the package; the other files are not redistributed because permission to redistribute them could not be confirmed (Table 1). Their download addresses and SHA-256 hashes are recorded so that the same files can be verified. Without them, a user can still rerun Study 2, check the archived results of Study 1 and rebuild every table and figure.

Table 1. Data sources

Data	Provider	Used in	Included
30 US industry portfolios, daily returns	Kenneth R. French Data Library	Study 1	No
Six equity indices, daily closing prices	Yahoo Finance	Study 1	No
Yen–US dollar exchange rate (DEXJPUS)	FRED, Federal Reserve Bank of St. Louis	Study 2	Yes

Source: *Data Sources*, Sections 1–3.

3. Computational Requirements

3.1 Software

The results were produced with 64-bit Python 3.11.9 on Windows 11. The file `requirements.txt` lists the 17 Python packages and their exact versions; the main packages are NumPy 2.4.3, pandas 3.0.1, SciPy 1.17.1, arch 8.0.0, statsmodels 0.14.6 and Matplotlib 3.11.0. R, Stata, LaTeX and Microsoft Word are not needed. The programs have not been tested on Linux or macOS.

3.2 Random numbers

The bootstrap tests use 9,999 draws. The random seed is 20261024. Study 2 uses this seed directly; in Study 1, each series uses the seed plus its position in the original order of the series.

3.3 Hardware and running time

The runs reported here used a Windows 11 computer with an Intel Core i9-12900H processor (20 logical processors) and 32 GB of memory. Study 1 runs in parallel with up to two fewer workers than the number of processors. The unzipped package needs about 61 MB of disk space. Table 2 lists the recorded running times.

Table 2. Running times

Step	Command	Time
Rerun Study 1 (36 series)	`reproduce.py study1`	1,571 s (26 min); 1,420 s (24 min) in a clean copy of the package
Rerun Study 2 (yen)	`reproduce.py study2`	145 s
Rebuild all tables and figures	`reproduce.py review`	21 s
Compare with the archived results	`reproduce.py compare`	less than 5 s

Source: `outputs/verification/final_execution_record.json` (Study 1). The other times were measured by the author in a separate rerun of Study 2 in a new Python environment.

4. Description of the Code

Table 3 describes the folders of the package. The folders in `code/core/` keep their original names because the modules import one another. The code is released under the MIT License (file `LICENSE`).

Table 3. Contents of the package

Folder or file	Content
`code/reproduce.py`	Main program. Runs each step in a new working folder and writes a log for each step
`code/data/`	Checks the hashes of the input files and copies the Study 1 files into the working folder
`code/core/`	Base models, FZ0 scores, MCS, the calibration methods and the Study 1 analysis
`code/pipeline/study2.py`	Study 2: reads the yen data and produces the forecasts and the test results
`code/reporting/`	Produces the tables and figures of the paper
`code/validation/`	Compares new results with the archived results and checks that each forecast uses only earlier data
`code/core/confirm/freeze_params.json`	Settings fixed before the evaluation, such as the bootstrap block lengths
`outputs/reference/`	Archived results reported in the paper
`outputs/verification/`	Records of the reproduction check
`provenance/`	Fixed settings, version-control records and the link between paper values and result files

5. Instructions for Replication

The steps below are for Windows PowerShell and are run from the unzipped package folder. Steps 1–3 and 5 need only the files in the package; Step 4 also needs the original Study 1 files.

Step 1. Install the software. Install 64-bit Python 3.11, create a virtual environment and install the packages in `requirements.txt`. In the later commands, `python` means the Python program of this environment (`.venv\Scripts\python.exe`).

```
py -3.11 -m venv .venv
.venv\Scripts\python.exe -m pip install -r requirements.txt
.venv\Scripts\python.exe -m pip check
```

Step 2. Check the package and create a working folder. The first command checks every file against `MANIFEST_SHA256.csv`. The second creates a new working folder, so that the archived results are never overwritten. The third checks the yen data; it reports that the Study 1 files are missing until Step 4.

```
python -I -B code/validation/check_integrity.py
python -I -B -X utf8 code/reproduce.py prepare --destination runs/r1
python -I -B -X utf8 runs/r1/code/reproduce.py inputs
```

Step 3. Rerun Study 2. This step estimates all models on the yen data, computes the forecasts of every method at the 97.5% and 99% levels, and repeats the bootstrap tests and the other analyses of Section 5.2 of the paper.

```
python -I -B -X utf8 runs/r1/code/reproduce.py study2
```

Step 4. Rerun Study 1 (optional). Place the French file and the six Yahoo files in a folder with the structure given in `data/ACCESS.md` (here `../study1`). The import accepts the files only if their hashes match, and the Study 1 run checks them again before estimation.

```
python -I -B -X utf8 runs/r1/code/data/manage_inputs.py import-study1 ../study1
python -I -B -X utf8 runs/r1/code/reproduce.py study1
python -I -B -X utf8 runs/r1/code/reproduce.py audit-study1
```

Step 5. Rebuild the tables and figures and compare the results. The first command produces the tables and figures; if Step 4 was skipped, the Study 1 tables are rebuilt from the archived Study 1 results. The second command changes future returns in temporary copies of the data and checks that earlier forecasts do not change. The third compares all new results with the archived results.

```
python -I -B -X utf8 runs/r1/code/reproduce.py review
python -I -B -X utf8 runs/r1/code/reproduce.py leakage
python -I -B -X utf8 runs/r1/code/reproduce.py compare
```

Each step writes a log to `outputs/verification/logs/` in the working folder and stops with a nonzero exit code if an error or a mismatch occurs.

6. List of Tables and Figures

Table 4. Programs that produce the tables and figures of the paper

Exhibit	Program	Output file
Tables 1–9 and C1–C4	`export_paper_tables.py`	`tables/Table_1.csv` to `tables/Table_C4.csv`
Figure 1	`paper_figures.py`	`figures/fig_tradeoff_en.pdf`
Figure 2	`paper_figures.py`	`figures/fig_cumulative_industries_en.pdf`
Figure 3	`paper_figures.py`	`figures/fig_cumulative_yen_en.pdf`
Figure 4	`paper_figures.py`	`figures/fig_miscalibration_gain_en.pdf`
Figure C1	`paper_figures.py`	`figures/fig_decades_en.pdf`

Note: The programs are in `code/reporting/`. Output files are in `outputs/reproduced/` of the working folder; each figure is also saved as PNG and SVG. `provenance/paper_mapping.md` links every table to the result files behind it.

7. Results of the Reproduction Check

The author reran both studies from the original input files with the settings fixed in advance. Table 5 compares the main results with the values in the paper.

Table 5. Main results: paper and rerun (97.5% level)

Comparison	Mean score difference	95% CI	*p*-value	Rerun versus paper
H1: EW+DJC − EW	−0.0079	[−0.0197, 0.0018]	0.0861	Identical
H2: EW+DJC − EW+BCd	−0.0208	[−0.0306, −0.0125]	0.0006	Identical
H3: EW+DJC − EW+YE	−0.0080	[−0.0160, −0.0017]	0.0478	Identical
Yen: EW+DJC − EW	−0.0071	[−0.0243, 0.0089]	0.1973	Identical

Note: H1–H3 use the 30 US industry portfolios; the yen row is the primary test of Study 2. The H1–H3 *p*-values are Holm-adjusted; the yen *p*-value is one-sided. The rerun values were compared at full precision. Source: `outputs/verification/headline_comparison.csv`.

In total, 427 numerical columns in the result files were compared: 415 are exactly equal, and the largest difference in the others is 1.3×10^{-15}. All 106 saved arrays of Study 2 are exactly equal. All 580 numbers in the 13 tables of the paper agree with the rerun, as do 49 further numbers quoted in the text, and the five figures were rebuilt from the saved results. In 38 further checks, returns after a given day were changed; the forecasts up to that day did not change. The package was also run in a new Python environment, in a folder outside the author's research files, and all steps finished without error.

The check used an earlier build of the paper whose tables and numbers are identical to the final version; the final version differs only in its title page, the wording of the Introduction, one added funding sentence and layout.

8. Limitations

1. The original daily VaR and ES forecasts of each Study 1 series were not saved. The rerun can therefore be compared with the saved daily panel scores and per-series statistics, but not with each original daily forecast.
2. Study 1 can be rerun exactly only with the original files, whose hashes are listed in *Data Sources*, Section 5. Files downloaded today may differ from those used in the paper.
3. The data are the current provider versions of historical series. They are not necessarily the data that were available in real time.
4. The periods of sharp yen appreciation and the signal groups in Section 5.2 use thresholds computed from the full sample. They describe the results after the fact and are not trading signals.
5. The order of events before each study was fixed is documented by the author's version-control records, summarized in `provenance/Prespecified_Design.md`.
6. The programs were run only on Windows. Results on other systems may differ slightly in the last digits.

9. References

Vilhuber, L., Connolly, M., Koren, M., Llull, J., and Morrow, P. (2022). A template README for social science replication packages, version 1.1. Social Science Data Editors (https://doi.org/10.5281/zenodo.7293838).